



pizza hut
pizza resto



PIZZA SALES ANALYSIS USING SQL



Developed by Vishakha



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ABOUT PIZZA

This project focuses on analyzing pizza sales data using SQL.

It includes data related to orders, pizza types, order details, pizzas.

The main goal is to understand sales performance and customer demand.

By using SQL queries, we identify top-selling pizzas, revenue trends , and ordering patterns to help improve business decisions.



PIZZA SALES EXECUTIVE DASHBOARD

Developed by Vishakha



\$817.9K

Total Revenue



21K

Total Orders



50K

Total Pizzas Sold



\$38.3

Avg Order Value

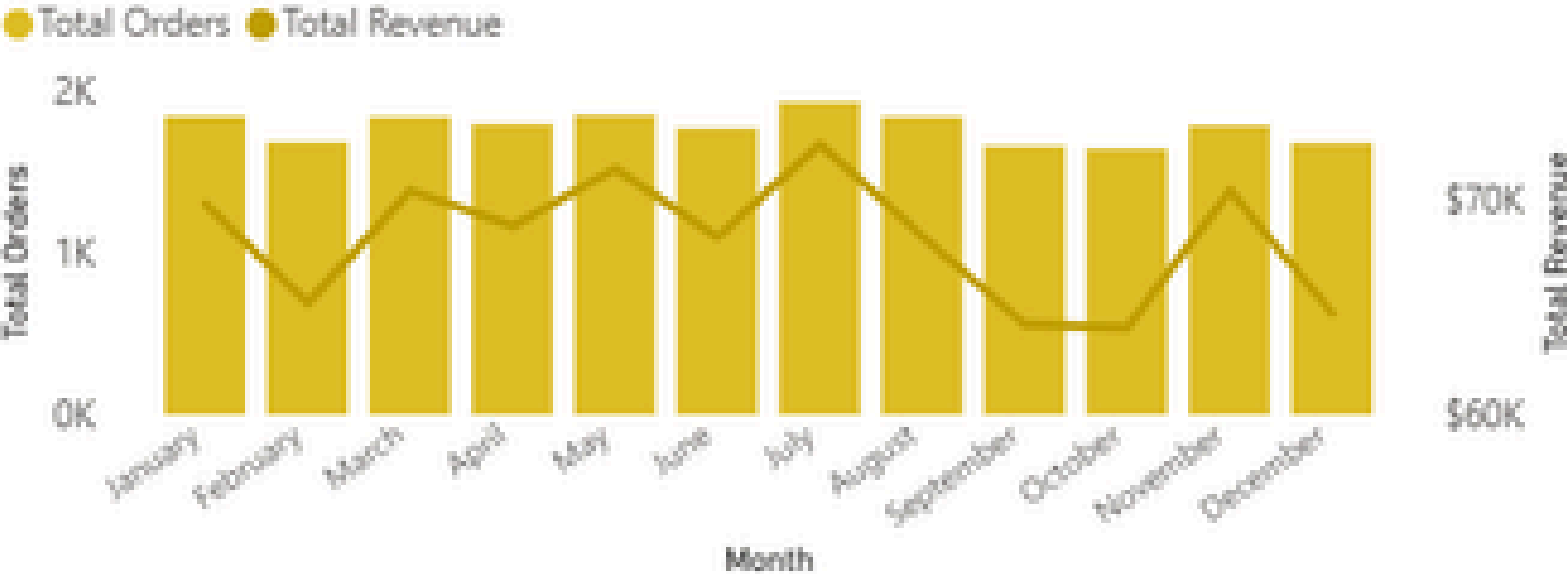
category

All

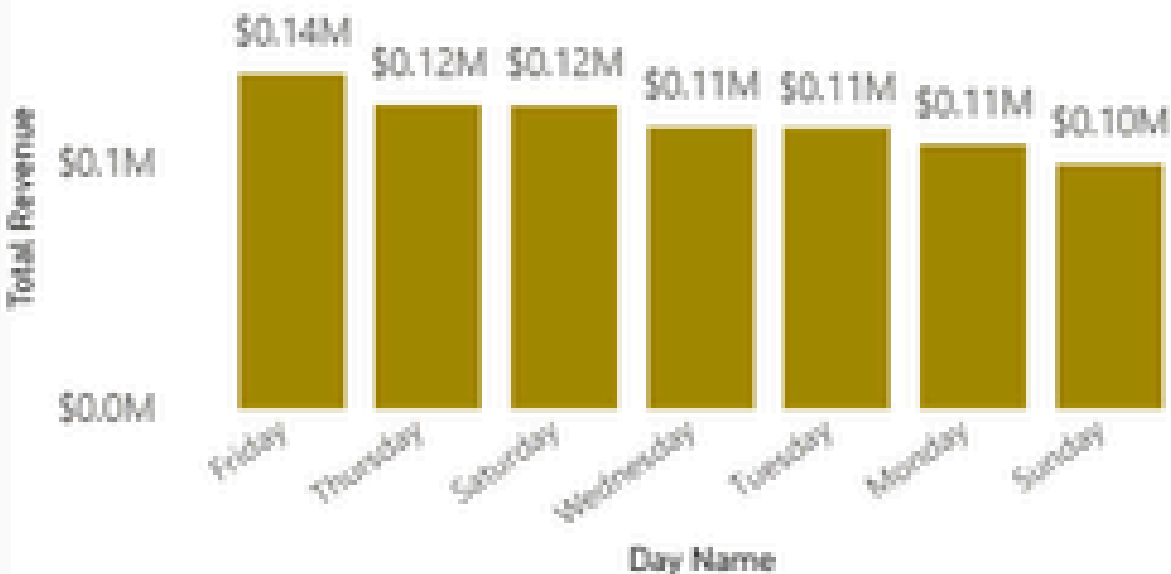
size

All

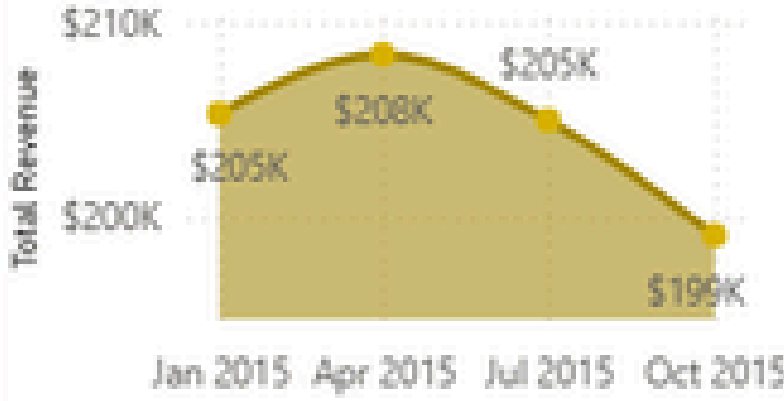
Total Orders and Total Revenue by Month



Total Revenue by Day Name



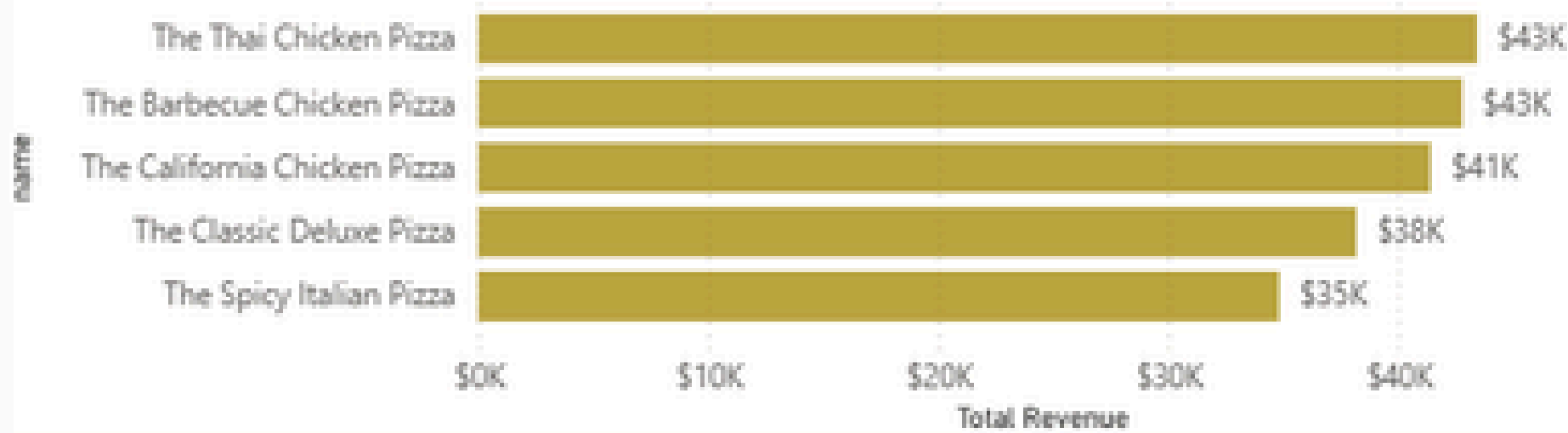
Total Revenue by Year and Quarter



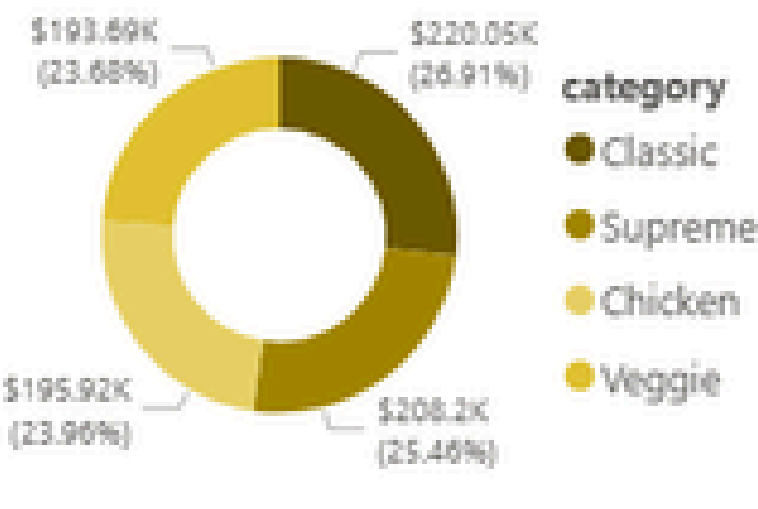
Total Orders by size



Total Revenue by name



Total Revenue by category





DATABASE OVERVIEW



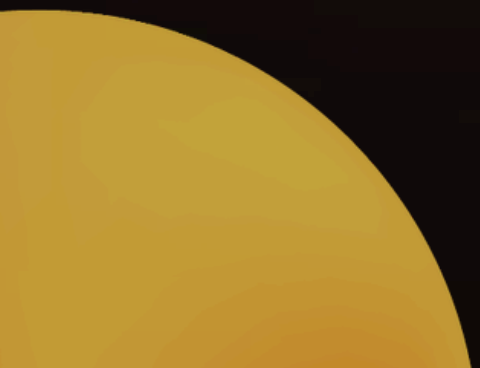
Database Name: pizza_sales

This database contains multiple tables used to store pizza sales data.

Tables are connected using relationships (Primary & Foreign Keys).

	Tables_in_pizza_sales
▶	order_details
	orders
	pizza_types
	pizzas

List of tables in pizza_sales database



DATA CLEANING



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- Converted date column into DATE format
 - Converted time column into TIME format
 - Changed data types to VARCHAR for consistency
 - Applied Primary Keys for unique identification
 - Applied Foreign Keys for table relationships
-
- **Data cleaning ensures accuracy, consistency, and proper relationships in the database.**

	Field	Type	Null	Key	Default
▶	order_id	int	NO	PRI	NULL
	date	date	YES		NULL
	time	time	YES		NULL

	Field	Type	Null	Key	Default
▶	order_details_id	bigint	NO	PRI	NULL
	order_id	int	YES	MUL	NULL
	pizza_id	varchar(20)	YES		NULL
	quantity	int	YES		NULL

Cleaned and structured table format



TOTAL ORDERS AND REVENUE



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Total Orders placed

```
• select
    count(order_details_id) as total_order
from
    order_details;
```

	total_order
▶	48620

#Total Revenue Generated

```
SELECT
    SUM( o.quantity * p.price ) AS total_pizza_revenue
FROM
    order_details o
inner JOIN
    pizzas p ON o.pizza_id = p.pizza_id;
```

	total_pizza_revenue
▶	817860.0499999993





COMMON AND HIGHEST PRICE PIZZA

#The highest-priced pizza.

```
SELECT Pt.Name, p.Price
FROM
    pizza_types pt
inner JOIN
    pizzas p ON pt.pizza_type_id = p.pizza_type_id
ORDER BY Price DESC
LIMIT 1;
```

	Name	Price
▶	The Greek Pizza	35.95

Most common pizza size ordered

```
SELECT Size, COUNT(*) AS OrderCount
FROM
    order_details od
inner JOIN
    pizzas p ON od.pizza_id = p.pizza_id
GROUP BY Size
ORDER BY OrderCount DESC
LIMIT 1;
```

	Size	OrderCount
▶	L	18526

MOST ORDERED PIZZA

List the top 5 most ordered pizza types along with their quantities.

```
SELECT pt.name,p.pizza_type_id, COUNT(*) AS Quantity
FROM
    pizzas p
inner join
    Pizza_types pt ON p.Pizza_type_id = pt.Pizza_type_id
inner JOIN
    order_details od ON od.pizza_id = p.pizza_id
GROUP BY pt.name, p.pizza_type_id
ORDER BY Quantity DESC
LIMIT 5;
```

	name	pizza_type_id	Quantity
▶	The Classic Deluxe Pizza	classic_dlx	2416
	The Barbecue Chicken Pizza	bbq_ckn	2372
	The Hawaiian Pizza	hawaiian	2370
	The Pepperoni Pizza	pepperoni	2369
	The Thai Chicken Pizza	thai_ckn	2315





DISTRIBUTION OF PIZZAS AND ORDERS

Distribution of orders by hour of the day.

```
SELECT
    CONCAT(HOUR(`time`), ' hour') AS order_hour,
    COUNT(order_id) AS total_orders
FROM orders
GROUP BY order_hour
ORDER BY order_hour;
```

	order_hour	total_orders
▶	10 hour	8
	11 hour	1231
	12 hour	2520
	13 hour	2455
	14 hour	1472
	15 hour	1468
	16 hour	1920
	17 hour	2336
	18 hour	2399

The category-wise distribution of pizzas.

```
SELECT
    pt.category,
    COUNT(p.pizza_id) AS total_pizzas
FROM
    pizzas p
JOIN
    pizza_types pt ON p.pizza_type_id = pt.pizza_type_id
GROUP BY pt.category
ORDER BY total_pizzas DESC;
```

	category	total_pizzas
▶	Veggie	27
	Classic	26
	Supreme	25
	Chicken	18



QUANTITY OF PIZZA

The total quantity ordered for each pizza category.

```
SELECT
    pt.category,
    SUM(od.quantity) AS total_quantity
FROM order_details od
inner JOIN pizzas p
    ON od.pizza_id = p.pizza_id
inner JOIN pizza_types pt
    ON p.pizza_type_id = pt.pizza_type_id
GROUP BY pt.category
ORDER BY total_quantity ASC;
```

	category	total_quantity
▶	Chicken	11050
	Veggie	11649
	Supreme	11987
	Classic	14888





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AVERAGE AND TOP ORDERED PIZZAS

The average number of pizzas ordered per day

```
SELECT
    AVG(daily_total) AS avg_pizzas_per_day
FROM (
    SELECT
        o.date,
        SUM(od.quantity) AS daily_total
    FROM orders o
    JOIN order_details od
        ON o.order_id = od.order_id
    GROUP BY o.date
) AS daily_orders;
```

	avg_pizzas_per_day
▶	138.4749

the top 3 most ordered pizza types based on revenue.

```
SELECT
    pt.name AS pizza_type,
    SUM(od.quantity * p.price) AS revenue
FROM order_details od
JOIN pizzas p ON od.pizza_id = p.pizza_id
JOIN pizza_types pt ON p.pizza_type_id = pt.pizza_type_id
GROUP BY pt.name
ORDER BY revenue DESC
LIMIT 3;
```

	pizza_type	revenue
▶	The Thai Chicken Pizza	43434.25
	The Barbecue Chicken Pizza	42768
	The California Chicken Pizza	41409.5



PERCENTAGE CONTRIBUTION OF PIZZA

The percentage contribution of each pizza type to total revenue.

```
SELECT
  pt.name AS pizza_name,
  CONCAT(
    ROUND(SUM(od.quantity * p.price) * 100.0 /
      (SELECT SUM(od2.quantity * p2.price)
        FROM order_details od2
        JOIN pizzas p2 ON od2.pizza_id = p2.pizza_id), 2
    ), '%'
  ) AS revenue_percentage
FROM order_details od
JOIN pizzas p ON od.pizza_id = p.pizza_id
JOIN pizza_types pt ON p.pizza_type_id = pt.pizza_type_id
GROUP BY pt.name
ORDER BY revenue_percentage DESC;
```

	pizza_name	revenue_percentage
►	The Thai Chicken Pizza	5.31%
	The Barbecue Chicken Pizza	5.23%
	The California Chicken Pizza	5.06%
	The Classic Deluxe Pizza	4.67%
	The Spicy Italian Pizza	4.26%
	The Southwest Chicken Pizza	4.24%
	The Italian Supreme Pizza	4.09%
	The Hawaiian Pizza	3.95%
	The Four Cheese Pizza	3.95%
	The Sicilian Pizza	3.78%
	The Pepperoni Pizza	3.69%
	The Greek Pizza	3.48%
	The Mexicana Pizza	3.27%

CUMULATIVE REVENUE

#The cumulative revenue generated over time.

```
SELECT
    daily.date,
    SUM(daily_revenue) OVER (ORDER BY daily.date) AS cumulative_revenue
FROM (
    SELECT
        o.date,
        SUM(od.quantity * p.price) AS daily_revenue
    FROM orders o
    JOIN order_details od ON o.order_id = od.order_id
    JOIN pizzas p ON od.pizza_id = p.pizza_id
    GROUP BY date
) AS daily
ORDER BY daily.date;
```

	date	cumulative_revenue
►	2015-01-01	2713.8500000000004
	2015-01-02	5445.75
	2015-01-03	8108.15
	2015-01-04	9863.6
	2015-01-05	11929.55
	2015-01-06	14358.5
	2015-01-07	16560.7
	2015-01-08	19399.05
	2015-01-09	21526.4
	2015-01-10	23990.350000000002
	2015-01-11	25862.65
	2015-01-12	27781.7
	2015-01-13	29831.300000000003



TOP ORDERED PIZZA

The top 3 most ordered pizza types based on revenue for each pizza category.

```
SELECT category, pizza_name, revenue
FROM (
  SELECT
    pt.category,
    pt.name AS pizza_name,
    SUM(od.quantity * p.price) AS revenue,
    RANK() OVER (
      PARTITION BY pt.category
      ORDER BY SUM(od.quantity * p.price) DESC
    ) AS rank_in_category
  FROM order_details od
  JOIN pizzas p ON od.pizza_id = p.pizza_id
  JOIN pizza_types pt ON p.pizza_type_id = pt.pizza_type_id
  GROUP BY pt.category, pt.name
) ranked
WHERE rank_in_category <= 3
ORDER BY category, rank_in_category;
```

	category	pizza_name	revenue
▶	Chicken	The Thai Chicken Pizza	43434.25
	Chicken	The Barbecue Chicken Pizza	42768
	Chicken	The California Chicken Pizza	41409.5
	Classic	The Classic Deluxe Pizza	38180.5
	Classic	The Hawaiian Pizza	32273.25
	Classic	The Pepperoni Pizza	30161.75
	Supreme	The Spicy Italian Pizza	34831.25
	Supreme	The Italian Supreme Pizza	33476.75
	Supreme	The Sicilian Pizza	30940.5
	Veggie	The Four Cheese Pizza	32265.70000000065
	Veggie	The Mexicana Pizza	26780.75
	Veggie	The Five Cheese Pizza	26066.5



**THANK YOU
FOR YOUR
ATTENTION**

